

Detailed Summary

“From-Scratch Neural Network for PII Detection” project was collaborated by Haley Banks, Kevelly Diaz, Nathaniel Boyle, and Hongyue Fu (April 2026) where they sketch out how to build a multi-layered neural network from scratch. The core objective of the technical project is to deploy this custom neural network for binary classification to identify whether business documents contain Personally Identifiable Information (PII).

Business Problem & Context

Organizations across public and private sectors routinely process vast volumes of consumer, employee, and vendor PII. Mishandling or exposing this data results in severe compliance, legal, and financial penalties. To emphasize the escalating scale of this issue, the authors highlight two major historical milestones:

- **Change Healthcare (2024):** A breach that exposed the sensitive PII of roughly 192.7 million individuals.
- **National Public Data (2024):** A massive security failure resulting in the leak of 2.9 billion records, which included Social Security numbers and deeply personal history profiles.

Core Conceptual Pillars

The instructional presentation is structured around four primary talking points:

1. **Data Engineering:** A breakdown of the selected document dataset, the rationale behind its choices, and the specific transformations applied to it.
2. **Text Representation:** An explanation of the vectorization and text embedding methodologies chosen to numerically represent the language features.
3. **Network Architecture & Theory:** A deep dive into the mathematical components of the custom network, the selected loss function, and the underlying mechanics of how a system "learns" via backpropagation.
4. **Execution Lifecycle:** A walk-through detailing the active training cycles, the mathematical production of optimal weight parameters, and model testing using those static configurations.

Live Demonstration & Corporate Utility

The group's demonstration plan involves executing one live training cycle consisting of 60 epochs, writing the generated weights to a saved file format, and running inference tests using those saved weights to categorize unseen text inputs.

The presentation concludes by framing the business value of this architecture. Constructing a private, in-house artificial intelligence engine provides a high-security automated layer to audit internal emails, corporate documents, and third-party files before transmission. This allows businesses to seamlessly flag or mask PII, drastically lowering operational risk and shielding the enterprise from compliance breaches and financial liabilities.

